Limits

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find $\lim_{x \to -1} (x^3 + 4x)$.

$= (-1)^3 + 4(-1) = -1 - 4 = -5.$

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find $\lim_{x \to \infty} \frac{3x^2 + 2x}{5x^2 - 1}$.

The numerator and denominator have the same degree (2), so by Theorem 6.1 the limit is the ratio of leading coefficients: $\frac{3}{5}$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

If $z_1 = 5 - 2i$ and $z_2 = -3 + 7i$, find the value of $2z_1 - z_2$ in standard form $a + bi$.

Step 1: Distribute scalar multiplication: $2z_1 = 2(5 - 2i) = 10 - 4i$ Step 2: Subtract z_2: $2z_1 - z_2 = (10 - 4i) - (-3 + 7i) = (10 - (-3)) + (-4 - 7)i = \mathbf{13 - 11i}$.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find $\lim_{x \to -9} \frac{x^2 + 14x + 45}{x + 9}$.

Final Answer: -4. Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 14x + 45}{x + 9} = \frac{(x + 9)(x + 5)}{x + 9} = x + 5$ (for $x \neq -9$), so the limit is $-9 + 5 = -4$.

5 **[Foundational]** • **Target: Direct substitution**

Find $\lim_{x \to 3} (2x^2 - x + 1)$.

$= 2(9) - 3 + 1 = 16.$

6 **[Foundational]** • **Target: Direct substitution**

Find $\lim_{x \to -1} (x^3 + 4x)$.

$= (-1)^3 + 4(-1) = -1 - 4 = -5.$

7 **[Foundational]** • **Target: Indeterminate forms: factor and cancel**

Find $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 16}{x - 4} = \frac{(x - 4)(x + 4)}{x - 4} = x + 4$ (for $x \neq 4$), so the limit is $4 + 4 = 8$.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Indeterminate forms: factor and cancel**

Find $\lim_{x \to -2} \frac{x^2 + 5x + 6}{x + 2}$.

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 5x + 6}{x + 2} = \frac{(x + 2)(x + 3)}{x + 2} = x + 3$ (for $x \neq -2$), so the limit is $-2 + 3 = 1$.

9 **[Intermediate]** • **Target: Indeterminate forms: factor and cancel**

Explain why you cannot find $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$ by direct substitution alone.

Substituting $x = 4$ gives $\frac{16 - 16}{4 - 4} = \frac{0}{0}$, an indeterminate form — it does not tell you the limit is 0 or undefined, so you must factor and cancel first.

10 **[Intermediate]** • **Target: Limits at infinity for rational functions**

Find $\lim_{x \to \infty} \frac{3x^2 + 2x}{5x^2 - 1}$.

The numerator and denominator have the same degree (2), so by Theorem 6.1 the limit is the ratio of leading coefficients: $\frac{3}{5}$.

11 **[Intermediate]** • **Target: Limits at infinity for rational functions**

Find $\lim_{x \to \infty} \frac{4x + 1}{2x^2 + 3}$.

The denominator has higher degree than the numerator, so by Theorem 6.1 the limit is 0.

12 **[Intermediate]** • **Target: Limits at infinity for rational functions**

Explain what happens to $\lim_{x \to \infty} \frac{5x^3 - 2}{x^2 + 1}$, without computing an exact value.

The numerator's degree (3) is higher than the denominator's (2), so by dividing by the highest power in the denominator the numerator's leading term still grows without bound — the limit does not exist as a finite number; the function increases without bound as $x \to \infty$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find $\lim_{x \to 3} (2x^2 - x + 1)$.

$= 2(9) - 3 + 1 = 16$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 16}{x - 4} = \frac{(x - 4)(x + 4)}{x - 4} = x + 4$ (for $x \neq 4$), so the limit is $4 + 4 = 8$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

© ClipSAT — clipsat.org | All Rights Reserved

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 5.01 Trigonometry & Polar Coordinates**

Find $\lim_{x \to -2} \frac{x^2 + 5x + 6}{x + 2}$.

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 5x + 6}{x + 2} = \frac{(x + 2)(x + 3)}{x + 2} = x + 3$ (for $x \neq -2$), so the limit is $-2 + 3 = 1$.

16 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 1.01 Complex Numbers**

Explain why you cannot find $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$ by direct substitution alone.

Substituting $x = 4$ gives $\frac{16 - 16}{4 - 4} = \frac{0}{0}$, an indeterminate form — it does not tell you the limit is 0 or undefined, so you must factor and cancel first.

17 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 2.01 Matrices & Vectors**

Find $\lim_{x \to \infty} \frac{4x + 1}{2x^2 + 3}$.

The denominator has higher degree than the numerator, so by Theorem 6.1 the limit is 0.

18 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 3.01 Sequences & Series**

Explain what happens to $\lim_{x \to \infty} \frac{5x^3 - 2}{x^2 + 1}$, without computing an exact value.

The numerator's degree (3) is higher than the denominator's (2), so by dividing by the highest power in the denominator the numerator's leading term still grows without bound — the limit does not exist as a finite number; the function increases without bound as $x \to \infty$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 4.01 Advanced Functions & Equations**

In triangle ABC, sides are $a = 5$, $b = 6$, and $c = 7$. Find the exact value of $\cos A$.

Law of Cosines solved for $\cos A$: $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ $\cos A = \frac{6^2 + 7^2 - 5^2}{2(6)(7)} = \frac{36 + 49 - 25}{84} = \frac{60}{84} = \mathbf{\frac{5}{7}}$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 5.01 Trigonometry & Polar Coordinates**

Let $f(x) = 3x - 1$ and $g(x) = 2x^2 + 5$. Find $f(g(3))$.

Step 1: Evaluate inner function $g(3)$: $g(3) = 2(3^2) + 5 = 2(9) + 5 = 18 + 5 = 23$. Step 2: Evaluate outer function $f(23)$: $f(23) = 3(23) - 1 = 69 - 1 = \mathbf{68}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find $\lim_{x \to 2} (-3x^2 + 7x - 7)$.

a. -3

b. -5 ✓

c. -7

d. 1

$= -3(2)^2 + 14 - 7 = -5$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find $\lim_{x \to 3} \frac{x^2 - 5x + 6}{x - 3}$.

a. 1 ✓

b. 3

c. -1

d. 4

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 5x + 6}{x - 3} = \frac{(x - 3)(x - 2)}{x - 3} = x - 2$ (for $x \neq 3$), so the limit is $3 - 2 = 1$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find $\lim_{x \to \infty} \frac{-9x^3 - 8x}{-8x^2 + 2x}$ (state whether it is ∞ or $-\infty$).

a. ∞ ✓

b. $\frac{9}{8}$

c. $-\infty$

d. 0

The numerator's degree (3) is higher than the denominator's (2), so the quotient behaves like $\frac{-9}{-8}x$ for large x. Since the leading coefficients -9 and -8 have same sign, the limit is ∞.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is $\lim_{x \to 0^-} \frac{1}{x}$?

a. $-\infty$ ✓

b. ∞

c. 0

d. Does not exist

As x approaches 0 through negative values, $\frac{1}{x}$ decreases without bound toward $-\infty$.

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find $\lim_{x \to -7} (3x^2)$.

a. 145

b. 147 ✓

c. -21

d. 149

$= 3(-7)^2 = 147.$

26 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find $\lim_{x \to -6} (2x^2 + 5x - 9)$.

a. 35

b. 31

c. -51

d. 33 ✓

$= 2(-6)^2 - 30 - 9 = 33.$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find $\lim_{x \to -7} (4x^2 - 2x - 9)$.

a. 203

b. 201 ✓

c. 199

d. -23

$= 4(-7)^2 + 14 - 9 = 201.$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find $\lim_{x \to -4} (3x^2 + x + 4)$.

a. -12

b. 50

c. 46

d. 48 ✓

$= 3(-4)^2 - 4 + 4 = 48.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find $\lim_{x\to 6}(2x^2+9x-6)$.

a. 122

b. 120 ✓

c. 60

d. 118

$=2(6)^2+54-6=120.$

30 [Intermediate] • Target: Direct substitution

Find $\lim_{x\to -3}(3x^2-8x-2)$.

a. 13

b. 51

c. 49 ✓

d. 47

$=3(-3)^2+24-2=49.$

31 [Intermediate] • Target: Indeterminate forms: factor and cancel

Find $\lim_{x\to -5}\frac{x^2-4x-45}{x+5}$.

a. 14

b. -16

c. -12

d. -14 ✓

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2-4x-45}{x+5}=\frac{(x+5)(x-9)}{x+5}=x-9$ (for $x\neq -5$), so the limit is $-5-9=-14$.

32 [Intermediate] • Target: Limits at infinity for rational functions

Find $\lim_{x\to -2}\frac{x^2+3x+2}{x+2}$.

a. 2

b. -1 ✓

c. -3

d. 1

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2+3x+2}{x+2}=\frac{(x+2)(x+1)}{x+2}=x+1$ (for $x\neq -2$), so the limit is $-2+1=-1$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find $\lim_{x\to -1}\frac{x^2-4x-5}{x+1}$.

a. -6 ✓

b. -8

c. 6

d. -4

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2-4x-5}{x+1}=\frac{(x+1)(x-5)}{x+1}=x-5$ (for $x\neq -1$), so the limit is $-1-5=-6$.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find $\lim_{x\to 7} \frac{x^2-16x+63}{x-7}$.

a. -2 ✓

b. -4

c. 0

d. 2

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2-16x+63}{x-7} = \frac{(x-7)(x-9)}{x-7} = x-9$ (for $x \neq 7$), so the limit is $7-9=-2$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find $\lim_{x\to -9} \frac{x^2+14x+45}{x+9}$.

a. -2

b. 4

c. -4 ✓

d. -6

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2+14x+45}{x+9} = \frac{(x+9)(x+5)}{x+9} = x+5$ (for $x \neq -9$), so the limit is $-9+5=-4$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find $\lim_{x\to \infty} \frac{-7x^2+3x}{-5x^2+4x}$.

a. $\frac{7}{5}$ ✓

b. $-\frac{7}{5}$

c. $\frac{2}{5}$

d. $\frac{12}{5}$

The numerator and denominator have the same degree (2), so the limit is the ratio of leading coefficients: $\frac{-7}{-5} = \frac{7}{5}$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find $\lim_{x\to 0} \frac{\sqrt{x+81}-\sqrt{81}}{x}$.

a. $\frac{1}{18}$ ✓

b. $\frac{1}{9}$

c. $\frac{1}{6}$

d. $\frac{2}{9}$

Multiply top and bottom by the conjugate $\sqrt{x+81}+\sqrt{81}$: the numerator becomes x, which cancels with the x in the denominator, leaving $1/(\sqrt{x+81}+\sqrt{81})$. As $x\to 0$: this is $1/(\sqrt{81}+\sqrt{81}) = 1/(2\sqrt{81}) = 1/(2\cdot 9) = \frac{1}{18}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Sequences & Series

Find $\lim_{x\to 0} \frac{\sqrt{x+25}-\sqrt{25}}{x}$.

a. $\frac{2}{5}$

b. $\frac{3}{10}$

c. $\frac{1}{10}$ ✓

d. $\frac{1}{5}$

Multiply top and bottom by the conjugate $\sqrt{x+25}+\sqrt{25}$: the numerator becomes x, which cancels with the x in the denominator, leaving $1/(\sqrt{x+25}+\sqrt{25})$. As $x\to 0$: this is $1/(\sqrt{25}+\sqrt{25}) = 1/(2\sqrt{25}) = 1/(2\cdot 5) = \frac{1}{10}$.

39 [Challenging] • Target: Direct substitution • Cross-Lesson: 4.01 Advanced Functions & Equations

Find $\lim_{x\to\infty} \frac{-9x^3-8x}{-8x^2+2x}$ (state whether it is ∞ or $-\infty$).

a. ∞ ✓

b. $\frac{9}{8}$

c. $-\infty$

d. 0

The numerator's degree (3) is higher than the denominator's (2), so the quotient behaves like $\frac{-9}{-8}x$ for large x. Since the leading coefficients -9 and -8 have same sign, the limit is ∞.

40 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Find $\lim_{x\to\infty} \frac{2x^3-8x}{9x^2-3x}$ (state whether it is ∞ or $-\infty$).

a. 0

b. $\frac{2}{9}$

c. $-\infty$

d. ∞ ✓

The numerator's degree (3) is higher than the denominator's (2), so the quotient behaves like $\frac{2}{9}x$ for large x. Since the leading coefficients 2 and 9 have same sign, the limit is ∞.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Complex Numbers

Find $\lim_{x\to 5} (-3x^2-9x+7)$.

a. -53

b. -113 ✓

c. -115

d. -111

$= -3(5)^2 - 45 + 7 = -113$.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Matrices & Vectors

Find $\lim_{x \to -4} (-2x^2 - 4x + 4)$.

a. -12 ✓

b. -10

c. -14

d. 28

$= -2(-4)^2 + 16 + 4 = -12$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Sequences & Series

Find $\lim_{x \to 3} \frac{x^2 - x - 6}{x - 3}$.

a. 7

b. 5 ✓

c. 3

d. -5

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - x - 6}{x - 3} = \frac{(x - 3)(x + 2)}{x - 3} = x + 2$ (for $x \neq 3$), so the limit is $3 + 2 = 5$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Functions & Equations

Find $\lim_{x \to 0} \frac{\sqrt{x + 25} - \sqrt{25}}{x}$.

a. $\frac{2}{5}$

b. $\frac{3}{10}$

c. $\frac{1}{10}$ ✓

d. $\frac{1}{5}$

Multiply top and bottom by the conjugate $\sqrt{x + 25} + \sqrt{25}$: the numerator becomes x, which cancels with the x in the denominator, leaving $1/(\sqrt{x + 25} + \sqrt{25})$. As $x \to 0$: this is $1/(\sqrt{25} + \sqrt{25}) = 1/(2\sqrt{25}) = 1/(2 \cdot 5) = \frac{1}{10}$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Evaluate: $\lim_{x \to 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2}$.

a. $-\frac{1}{4}$ ✓

b. $\frac{1}{4}$

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

Combine numerator fractions: $\frac{2 - x}{2x(x - 2)} = \frac{-(x - 2)}{2x(x - 2)} = -\frac{1}{2x}$. As $x \to 2$, this equals $-\frac{1}{4}$.

© ClipSAT — clipsat.org | All Rights Reserved